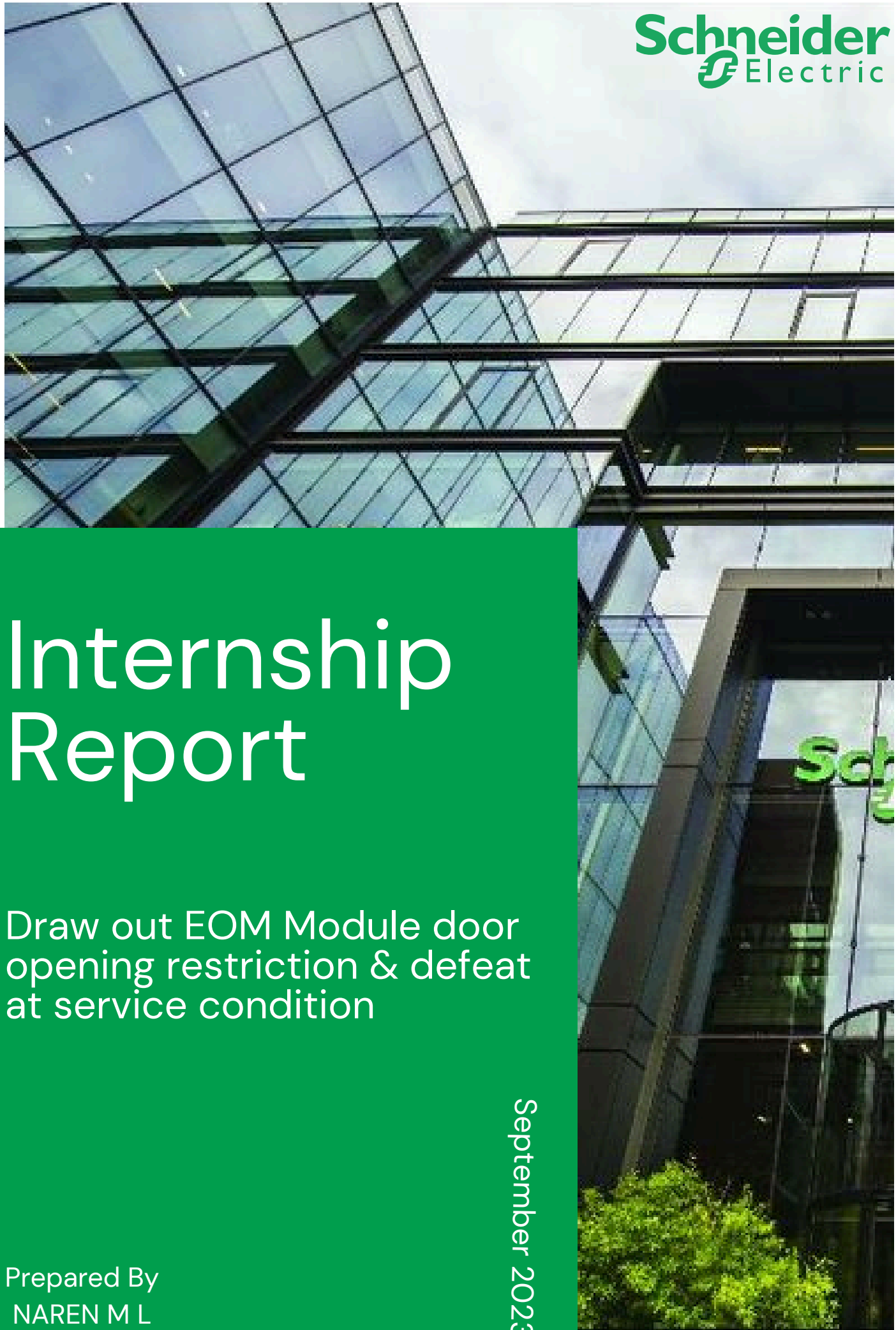


Internship Report

Draw out EOM Module door opening restriction & defeat at service condition

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Table of Contents

About Schneider Electric	3
Problem statement	4
Getting started	7
Solution	8
Idea generation	9
Concept finalization	10
Preliminary design	11
Conclusion	13

About Schneider Electric

Schneider Electric, founded in 1836, is a global leader in energy management and automation solutions. Operating in over 100 countries, they offer a wide range of innovative products and software for various sectors, leveraging IoT and AI through their EcoStruxure™ platform. They're committed to advancing energy efficiency and sustainability, helping organizations make data-driven decisions while emphasizing corporate responsibility. Schneider Electric's extensive product portfolio spans electrical distribution, industrial automation, building management systems, renewable energy solutions, and more. With a strong financial standing and a focus on innovation and sustainability, they're poised to shape the future of energy management and automation worldwide.

My role

I worked at Schneider Electric in the Switchgear Design and Development Center, where I was actively involved in various aspects of designing and developing switchgear products. Schneider Electric is a globally renowned company specializing in energy management and automation solutions, and its Switchgear Design and Development Center is at the forefront of innovation in the electrical industry.



Figure showing a Low voltage Switchboard

Problem statement

Aim

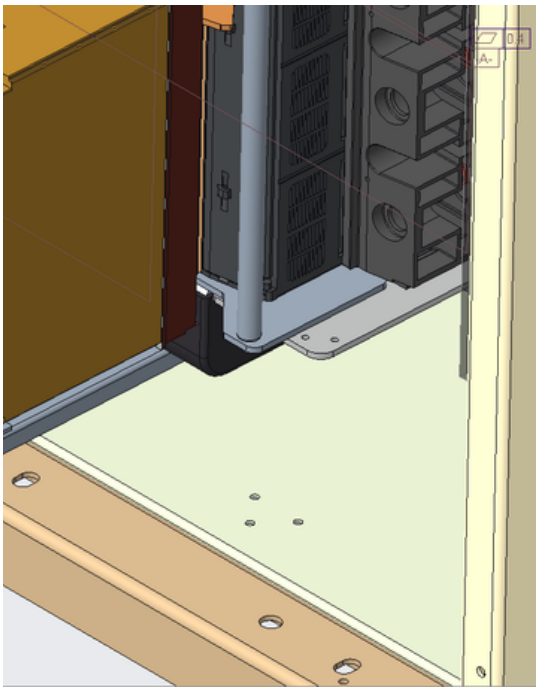
To design and develop a door interlock system for draw out EOM Module door with defeat feature.

Objectives

- User shouldnt be able to open the door untill isolated position is reached.
- On well aware conscious state, the user should be able to bypass the system(defeat feature) and open the door at test position.
- If done so , the door interlock system should be able to lock itself up, by only closing the door.
- The system should faciltate the complete draw out feature of EOM Module, allowing it to completely come out of the cradle, without causing any disturbance to the motion of the breaker.
- After putting back the drawn out breaker, the user should not be in a position to reset any feature,i.e, the door interlock system, should be capable of resetting itslef in the case of closing the door after putting back the EOM module.

Scope

- It is clear that, the system that is supposed to be achieves multiple requirements as a single project.
- The accuracy the system is very crucial, as we are dealing with safety parametrs here.
- It should capable of withstanding unregulated amounts of stresses, as it functions as a door lock against a human user.



Space available for the Door interlock system

Getting started

The whole engineering process of designing and developing the system went strictly by the principles of 'Design Thinking', which made this effort a successfull one.

What is Design thinking ?

Design thinking is a flexible and iterative process that encourages a user-centric approach to problem-solving. It is widely used in various industries to develop products, services, and solutions that are more innovative, user-friendly, and aligned with the needs of the target audience. It typically involves a five-step process:

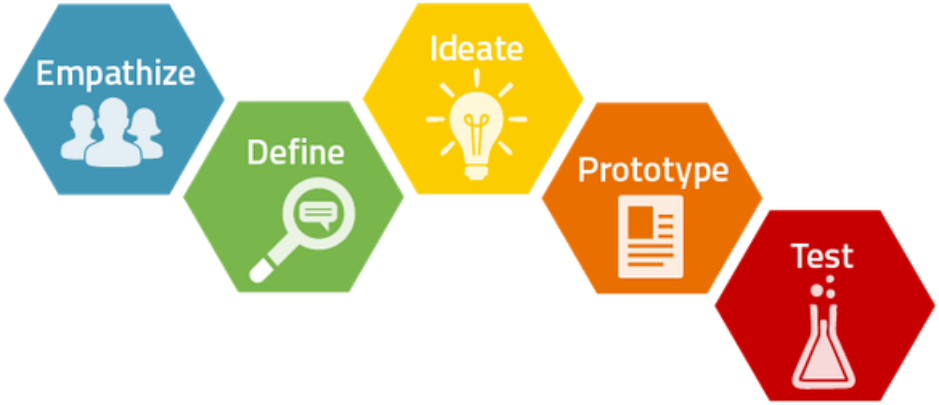
1. **Empathize**: In this initial stage, designers seek to understand the problem from the perspective of the people who are experiencing it. They engage in active listening, conduct interviews, and immerse themselves in the users' experiences to gain insights into their needs, challenges, and emotions.

2. **Define**: Once designers have gathered insights, they define the problem statement in a clear and actionable way. This step helps to narrow down the focus and ensures that the team is addressing the right issues. It often involves synthesizing information from the empathy phase to create a specific problem statement.

3. **Ideate**: During the ideation phase, participants generate a wide range of ideas to solve the defined problem. This step encourages brainstorming, free thinking, and creativity without judgment. The goal is to explore as many potential solutions as possible.

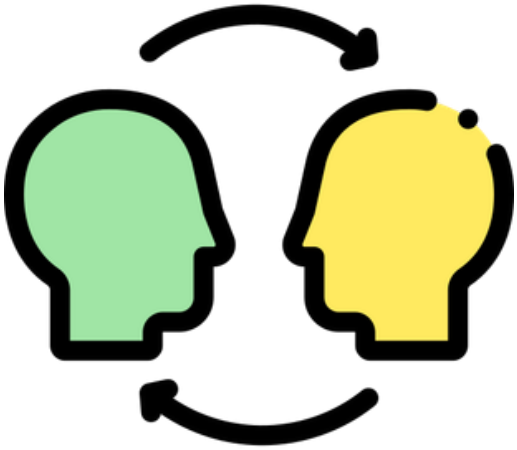
4. **Prototype**: Prototyping involves creating a tangible representation of one or more of the ideas generated during the ideation phase. Prototypes can be sketches, physical models, digital mock-ups, or any other form that allows for testing and experimentation.

5. **Test**: In the final stage, designers gather feedback on their prototypes by testing them with users. This feedback loop helps refine and improve the solutions, ensuring that they meet users' needs and expectations.



Solution

Empathize



- Knowing, how user operates the racking process, helped a lot to understand the problem that is needed to be solved.
- There was a need for a robust system, with minimum possibilities of failure and capable of chieveing multiple objectives.
- Based omn the fact, that it is Electrically Operated Module, (EOM) the cahnces of manual resetting minimiseze to zero.
- The interlock should difficult to bypass from unautherised users, while authorised users should be comfortable with it.

Definintion:



- With all these facts known, it was easy to define what users need from the system and what they dont want from it.
- The system is a safety oriented mechansim.It prevents users from getting in contact with live wires and bsusbars in service condition, by unintentional opening of door.
- the system should not disturb the draw out feature of the module, while providing its purpose.

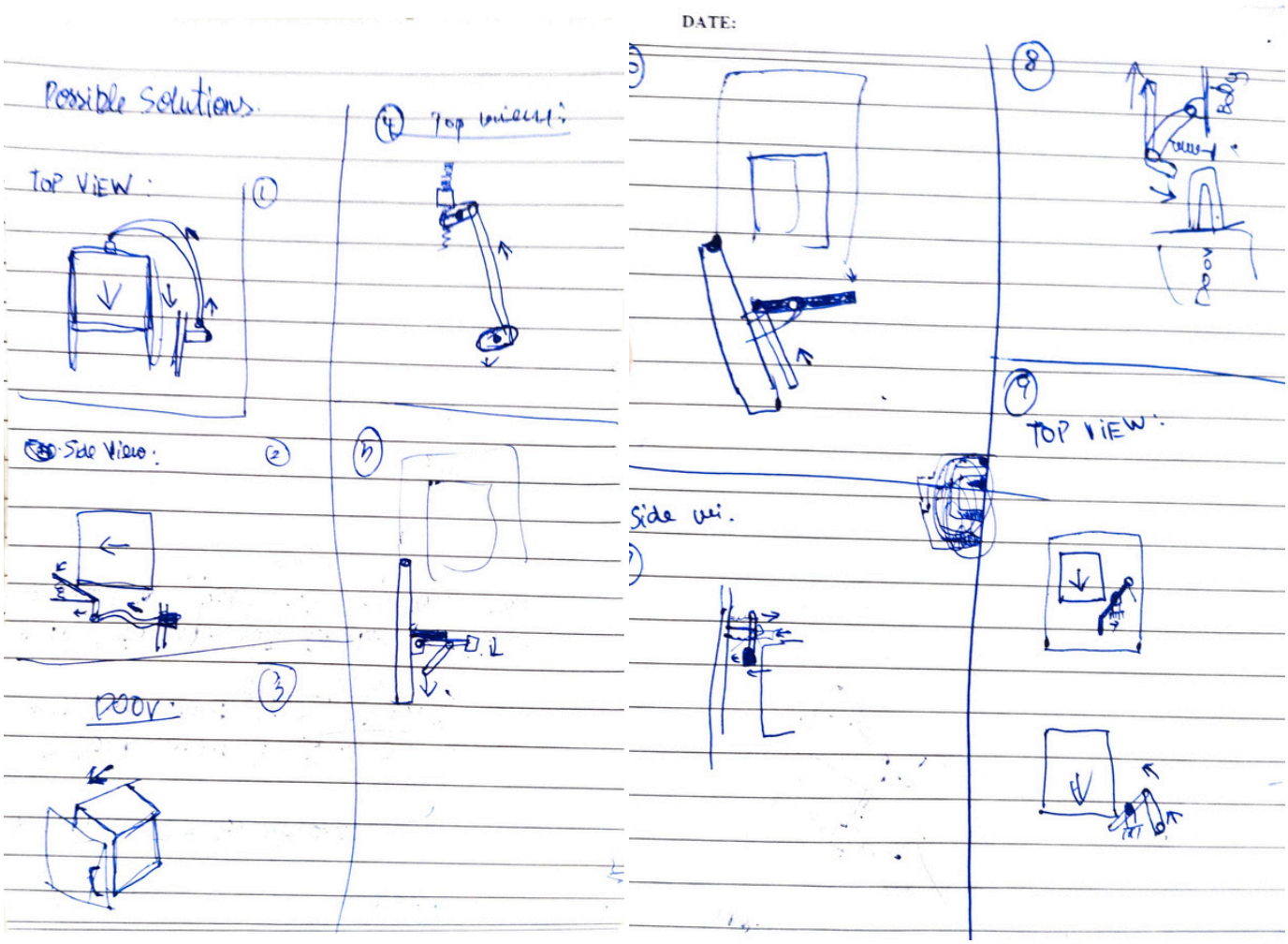
Idea Generation

Idea Screening

All ideas are brainstormed and encoureged to be documented for the idea screening process. this is where ideas are scored against ecah oither and ideas that does not meet objectives,unfeasible, or inpractical arte elimnated.

Many number of possible solutions were conceptualised.

Rough sketches:



Problems encountered :

- there was very little space to work and implement the system.
- The interlock system had to only rely on the movement of the breaker as there were no other moving part.

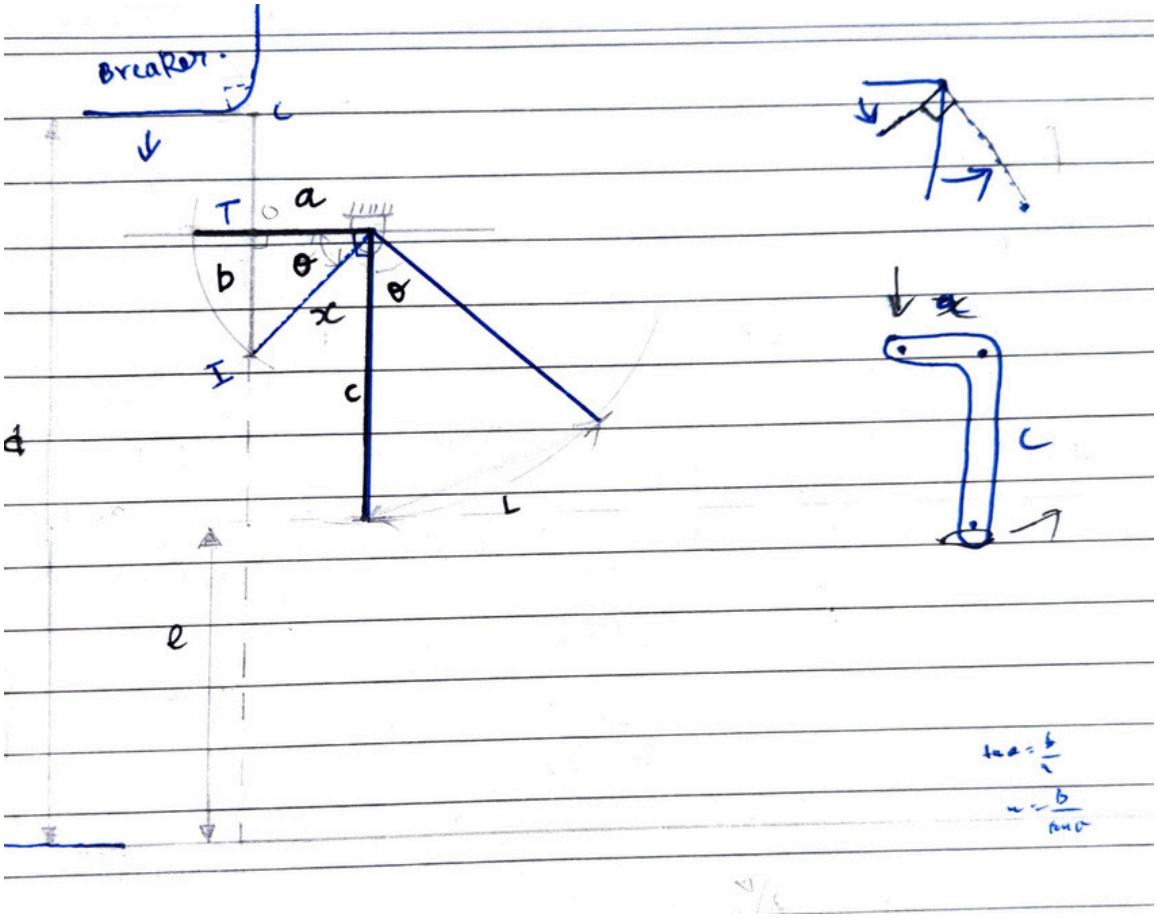
Concept finalisation

Out of all concepts, one concept is selected for its distuinguised capabilituies and further developed into a product.

About the concept

- The idea is to design an L shaped structure with vertical flat poles at either end of the “L” structure.
- While one end lies in the path of the breaker,it will get pushed by the breaker causing an opposite reaction at the other end.
- Since the “L” shaped structure is pivoted in the midle,One end acts as a trigger to transfer the motion of the breaker to the other end which actually is the lock for the door.

The selected concept was further refined usin quantifieble values and calculations were performed to give accurate results.



Door Interlock and Defeat mechanism- Calculations

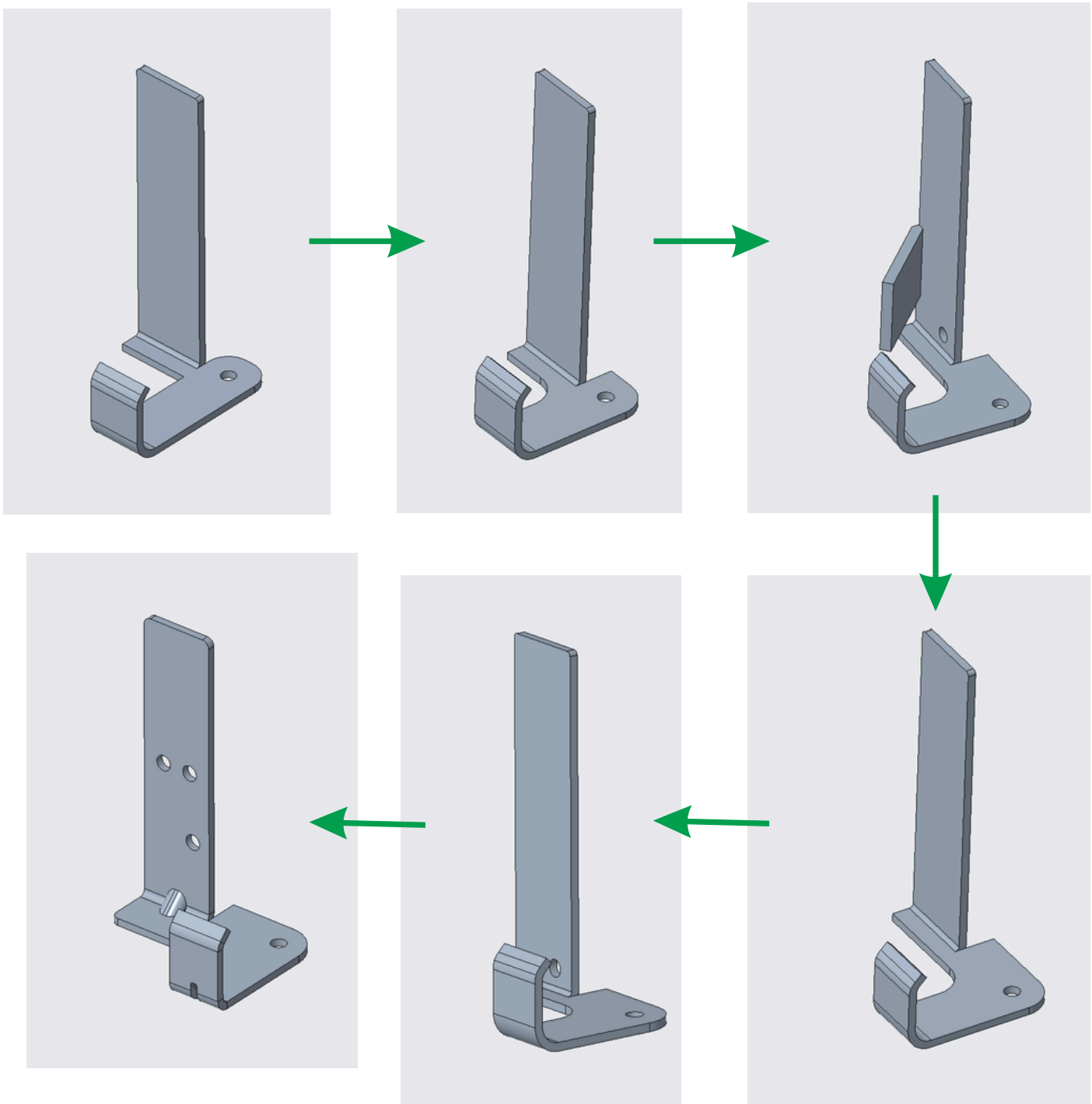
	INPUT PARAMETERS		RESULTS	
constants	d(mm)	113.19	theta(θ)	30
	e(mm)	60.11	c(mm)	33.08
	b(mm)	10		
			a(mm)	17.32051
	x(mm)	20	L(mm)	17.12347

Door Interlock and Defeat mechanism- Calculations

	INPUT PARAMETERS		RESULTS		
					corrected(θ)
constants	d(mm)	113.19	theta(θ)	56.30993	53.30993
	e(mm)	60.11	c(mm)	33.08	
	b(mm)	30			
			X_{min}(mm)	37.41212	
	a(mm)	20	L(mm)	31.21812	
				(min=7.5)	

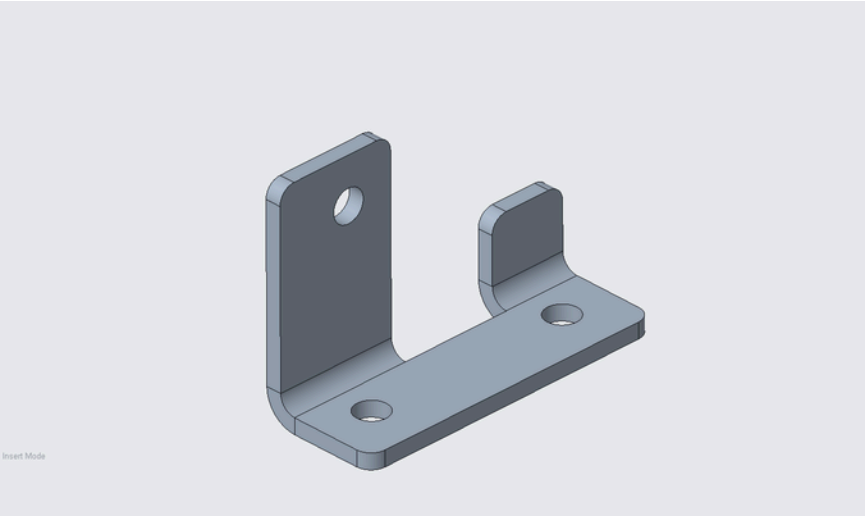
Preliminary Design

Multiple designs were made and iterations were performed to get the final desired output.

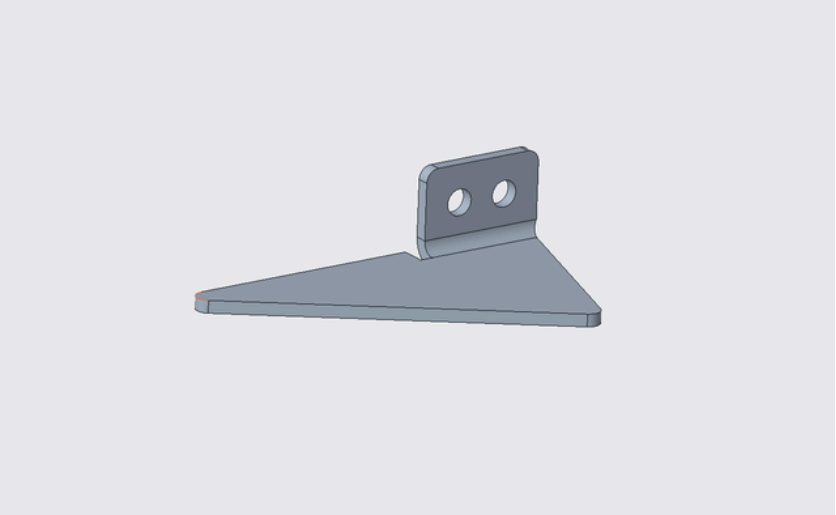


Final Designs

In similar pattern, rest of the two parts were designed



The part responsible for holding the movable arm



The part responsible for resetting the system

Conclusion

During my internship at Schneider Electric's Switchgear Design and Development Center, I worked on a project to create a Door Interlock System for the draw-out EOM Module door. This system aimed to enhance safety by preventing the door from opening until the isolated position was reached, while also allowing for a temporary defeat feature for testing purposes. When the door was opened in this state, it would self-lock upon closure. Additionally, the system enabled a smooth draw-out of the EOM Module and prevented users from resetting any features after reinsertion.

The project's scope was extensive, addressing multiple requirements, with safety and precision as top priorities. The Door Interlock System had to withstand substantial stress levels, serving as a robust door lock against users. This project showcased Schneider Electric's commitment to innovative solutions, emphasizing safety and precision.

In summary, my internship at Schneider Electric allowed me to contribute to the development of a highly effective Door Interlock System, aligning with Schneider Electric's dedication to safety and technological advancement in the electrical industry.

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